# Practice Set 4 Paper 1 – Basic Sciences in Urology

**Contents**


1. Q1. Pathophysiology of renal stones and prevention of recurrence (LAQ)
2. Q2. Pathophysiology of OAB; outline of management; non-pharmacological management (LAQ)
3. Q3a. Xanthogranulomatous pyelonephritis (SAQ)
4. Q3b. Doppler in urology (SAQ)
5. Q3c. Christian Chaussey (SAQ)
6. Q3d. Creatinine clearance (SAQ)
7. Q3e. Radionuclear studies in PUJ obstruction (SAQ)
8. Q3f. Cyclosporins (SAQ)


*Practice paper compiled from the question bank.*

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## Q1. Discuss pathophysiology of renal stones and also describe how to prevent its recurrence

**Type:** LAQ (25 marks) (also answered in 2020-Summer Paper 4, Q2)

### Introduction

- Stone disease is a **chronic metabolic disease** with recurrence up to **50% within 5 years**. Removal is surgical; recurrence is metabolic.
- Calcium oxalate (with or without apatite) forms about 80%; uric acid and struvite about 7% each; cystine 1–3%.

### Risk factors

- **Epidemiological:** male sex, hot climate (low urine volume), animal protein and salt, family history.
- **Systemic:** metabolic syndrome and diabetes (insulin resistance lowers urine pH), gout, hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass, sarcoidosis.
- **Stasis:** PUJ (pelviureteric junction) obstruction, horseshoe kidney, medullary sponge kidney. **Drugs:** indinavir, topiramate, vitamin C and D excess.

### Physicochemistry: state of saturation

| State | Concentration product (CP) | What happens |
|---|---|---|
| Undersaturated | Below Ksp (solubility product) | No crystals; stones may dissolve |
| **Metastable** | Between Ksp and Kf (formation product) | No spontaneous nucleation, but growth on existing crystals; inhibitors act here. **Zone of therapy** |
| Unstable | Above Kf | Spontaneous nucleation, growth, aggregation |

### Steps of stone formation

1. **Supersaturation:** low volume; high calcium, oxalate or uric acid; low citrate.
2. **Nucleation:** homogeneous, or **heterogeneous** on debris or other crystals (needs less energy).
3. **Growth and aggregation:** aggregates, not single crystals, can block a tubule within the 5–7 minute transit (**free-particle theory**).
4. **Retention (fixed-particle theory): Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) point to a secondary cause: enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria.

### Inhibitors and matrix

- **Inhibitors: citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm–Horsfall protein**, osteopontin. **No inhibitor of uric acid** exists. Matrix is about 2.5% of stone weight.

### Pathophysiology by urinary abnormality

| Abnormality (definition) | Mechanism |
|---|---|
| **Hypercalciuria** (>200 mg/day, or >4 mg/kg/day) | Absorptive (gut), renal leak (secondary high PTH [parathyroid hormone]), resorptive (primary hyperparathyroidism): Pak classification |

| Abnormality (definition) | Mechanism |
|---|---|
| Hyperoxaluria (>40 mg/day) | **Enteric:** malabsorbed fat binds calcium, freeing oxalate; **primary:** hepatic enzyme defect (PH1, AGXT gene) |
| Hyperuricosuria (>600 mg/day) | Urate promotes calcium oxalate nucleation |
| **Hypocitraturia** (<320 mg/day) | Distal RTA, diarrhoea, hypokalaemia, animal protein (acid load) |
| **Uric acid stones** | **Low urine pH (<5.5)** more than excess uric acid; pKa 5.5 |
| **Cystine** | Autosomal recessive defect of tubular cystine reabsorption |
| **Struvite** | **Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate and carbonate apatite |

## Prevention of recurrence

### 1. Screen every first-time former (AUA [American Urological Association], EAU [European Association of Urology])

- History, serum electrolytes, creatinine, calcium, uric acid; urine pH and culture; **stone analysis**; imaging. This sorts patients into high and low risk.

### 2. General measures (all formers)

- **Fluid for urine ≥2.5 L/day** (drink about 3 L). Borghi 1996: 5-year recurrence 12.1% vs 27%.
- **Normal calcium**, low sodium, less animal protein, more fruit and vegetables (Borghi 2002: halved recurrence versus a low-calcium diet, which raises oxalate absorption).
- Limit high-oxalate foods (spinach, nuts, chocolate); weight loss.

### 3. Metabolic evaluation (high-risk or recurrent formers, children)

- One or two **24-hour urines**: volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine. Therapy targeted to these results reduces stone events.

### 4. Selective drug therapy

| Abnormality | Treatment | Rationale |
|---|---|---|
| Hypercalciuria | **Thiazide:** hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg | Distal calcium reabsorption; add potassium citrate (hypokalaemia causes hypocitraturia) |
| Hypocitraturia, distal RTA | **Potassium citrate 30–60 mEq/day** | Alkali load reduces tubular citrate reabsorption |
| Hyperuricosuria with calcium stones | Allopurinol 300 mg/day | Watch hypersensitivity |
| Uric acid stones | Potassium citrate to **pH 6.0–6.5** (avoid >7.0) | pH disease; allopurinol only for hyperuricosuria |
| Enteric hyperoxaluria | **Calcium with meals**, low-oxalate, low-fat diet, magnesium, liquid citrate | Binds oxalate in gut |
| Primary hyperoxaluria type 1 | Pyridoxine; **lumasiran** (RNA [ribonucleic acid] interference) | ILLUMINATE-A |
| Cystinuria | Urine ≥3 L/day, pH **7.0–7.5**; **tiopronin** | Cystine <250 mg/L; thiol forms soluble mixed disulfide |
| Struvite | **Complete clearance**, antibiotics; AHA (acetohydroxamic acid) 250 mg three times daily if unfit for surgery | Urease inhibitor; thrombosis |

**5. Monitoring**

- 24-hour urine within **6 months** of starting therapy, then yearly; reanalyse any new stone.

## Recent advances

- **ROKS (Recurrence of Kidney Stone) nomogram** predicts individual risk. **Also know:** NOSTONE (2023): hydrochlorothiazide did not reduce recurrence; citrus juices are adjuncts; *Oxalobacter formigenes* loss.

**Diagram to draw:** saturation zones (Ksp, Kf) and Randall plaque on a renal papilla.

> **Key points to remember:**
>
> - Metastable zone between Ksp and Kf: where inhibitors and therapy act.
> - Randall plaque anchors idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
> - Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi).
> - Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.

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## Q2. Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management

**Type:** LAQ (25 marks) (neuromodulation also answered in 2025-Winter Paper 3, Q3e and 2023-Winter Paper 2, Q3b)

### Definition

- **OAB (overactive bladder), ICS (International Continence Society):** urgency, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), without UTI (urinary tract infection) or other obvious pathology.
- OAB is a **symptom** diagnosis. DO (detrusor overactivity) is a **urodynamic** finding; only 44–69% of patients with urgency show DO.

### Pathophysiology

All mechanisms converge on **more afferent signal reaching awareness**: the bladder turns the signal up, or the CNS (central nervous system) fails to turn it down.

| Mechanism | Explanation |
|---|---|
| **Afferent (sensory)** | Aδ fibres normally sense filling. Unmyelinated **C fibres** are "silent", but become spontaneously active and hypersensitive in pathological states |
| **Urothelial** | Urothelium acts as a transducer: mechanical, inflammatory or chemical stimuli release ATP (adenosine triphosphate), acetylcholine and other mediators onto suburothelial afferents |
| **Myogenic** | Patchy denervation causes receptor upregulation and spontaneous contraction; exaggerated **micromotions** cause urgency, and coordinated spread across the wall gives DO |
| **Neurogenic / central** | Forebrain and PAG–PMC (periaqueductal grey–pontine micturition centre) normally inhibit voiding; loss of this inhibition lets "afferent noise" be felt as urgency |
| **Other** | Bladder–bowel crosstalk (constipation), ischaemia–reperfusion, chronic inflammation, altered urinary microbiome |

- **Risk factors:** ageing, female sex, BOO (bladder outlet obstruction: an animal model, but only an indirect cause in humans; DO persists in up to 60% after TURP [transurethral resection of the prostate]), anxiety and depression, metabolic syndrome, poor toileting habits.

### Brief outline of management

- **Assessment:** history, examination, **urinalysis** (minimum); bladder diary (3–7 days); questionnaires; PVR (post-void residual) when indicated. Urodynamics only if it will change an invasive decision.
- **Ladder (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]):** not a compulsory sequence.

```
Education + behavioural therapy (first line)
        ↓  (4–12 weeks)
Antimuscarinic or β3-agonist (mirabegron, vibegron)
        ↓
PTNS / onabotulinumtoxinA / SNM (third line)
        ↓
Augmentation cystoplasty or urinary diversion
```

- Antimuscarinics: dry mouth, constipation; **dementia risk with use >3 months**, so β3-agonists are preferred in the elderly.

## Non-pharmacological management

**1. Education and lifestyle**

- Explain normal bladder function; set realistic goals.
- Reduce fluid intake by about 25% if excessive; cut caffeine; treat constipation.
- **Weight loss:** about 8 kg loss in obese women reduced UUI episodes.

**2. Bladder training (BT)** (breaks the cycle of urgency, frequent voiding, smaller capacity)

1. Keep a bladder diary.
2. Start a fixed interval (often 1 hour while awake).
3. Void by the clock, not by urge.
4. Increase by 15–30 minutes a week as control improves.
5. Target interval 3–4 hours.

- Works by **uncoupling the void from the urge**.

**3. PFMT (pelvic floor muscle training) and urge suppression**

- Kegel exercises: quick 1–2 s flicks and 5–10 s holds, twice daily.
- A pelvic floor contraction **reflexly inhibits the detrusor**. Urge suppression: stop, stay still, squeeze quickly several times, relax, let the urge pass, then walk calmly to the toilet.
- PFMT and BT reduce incontinence episodes by **50–80%**.
- **Also know:** biofeedback helps patients who cannot find the muscle; electrical stimulation at 5–20 Hz inhibits the detrusor; timed and prompted voiding for cognitively impaired patients.

**4. PTNS (percutaneous tibial nerve stimulation)**

- 34-gauge needle 3–4 cm above the medial malleolus; 20 Hz, 200 μs; confirm great-toe flexion or sole tingling. **30 min weekly for 12 weeks**, then maintenance.
- Posterior tibial nerve (L4–S3) shares sacral roots with the bladder. **SUmiT trial:** 55% vs 21% (sham) responders.

**5. SNM (sacral neuromodulation)**

- Mechanism: stimulates **afferents** that modulate brainstem and forebrain control.
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or **staged tined lead** at **S3** (anal "bellows" and great-toe flexion). ≥50% improvement on diary → implant pulse generator in the buttock.
- **InSite trial:** 61% vs 42% (standard medical therapy). Revision >30% at 5 years; MRI (magnetic resonance imaging)-conditional rechargeable devices now available.

**6. Intradetrusor onabotulinumtoxinA**

- Cleaves SNAP-25 (synaptosomal-associated protein 25), blocking acetylcholine release; also reduces afferent signalling.
- **100 U in 10 mL at 20 sites** cystoscopically; effect lasts about 6–9 months.
- About 25% become completely dry. Risks: UTI about 20%, retention needing CISC (clean intermittent self-catheterisation). **ROSETTA:** greater UUI reduction than SNM at 6 months, but more UTI.

**7. Last resort:** augmentation cystoplasty or urinary diversion.

**Diagram to draw:** bladder showing urothelial afferents, C fibres and micromotions, with the brainstem and forebrain inhibitory pathways.

**Key points to remember:**

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- BT and PFMT with urge suppression first: 50–80% fewer leaks.
- PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

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## Q3a. Xanthogranulomatous pyelonephritis.

**Type:** SAQ (10 marks)

### Definition

- XGP (xanthogranulomatous pyelonephritis) is a rare, severe, **chronic** infection causing diffuse destruction of an enlarged, non-functioning kidney, with **lipid-laden foamy macrophages** (xanthoma cells).
- Up to a fifth of nephrectomies for pyelonephritis. Peak age 50–70 years; women; diabetes is a risk factor.

### Pathogenesis

- **Triad: stone, obstruction, infection.** Stones in up to 80% (half staghorn).
- Low-virulence organisms (**Proteus** commonest, *E. coli*) in an obstructed, ischaemic kidney with an inadequate acute inflammatory response.

### Pathology

- Diffuse (>80%) or segmental (focal).
- Gross: yellow nodules, pus-filled dilated calyces, a **contracted pelvis** encased by fibrosis, frequent **perinephric extension**.
- Microscopy: sheets of foamy histiocytes with lymphocytes, plasma cells and giant cells.
- **Malek staging:** I confined to kidney; II into perinephric fat; III into paranephric tissue or retroperitoneum.

### Clinical features

- Flank pain, fever (about two-thirds), flank mass (about 60%), weight loss; persistent bacteriuria.
- Labs: anaemia, raised inflammatory markers, pyuria; liver dysfunction in up to half. One-third have **sterile urine** (prior antibiotics).

### Imaging

- **CT (computed tomography), most useful:** enlarged kidney, little or no function, central calculus in a contracted pelvis; low-density cavities with rim enhancement (**"bear paw" sign**); extrarenal spread.
- Ultrasound: enlarged kidney with hypoechoic areas. DMSA (dimercaptosuccinic acid) scan quantifies lost function.
- **Differentials:** RCC (renal cell carcinoma), pyonephrosis, malakoplakia, lymphoma, tuberculosis.

### Management

- Antibiotics to stabilise; drain if septic.
- **Nephrectomy** (with perinephric fat) for diffuse disease; partial nephrectomy for focal disease.
- Surgery is difficult: dense adhesions to psoas, great vessels, bowel and diaphragm. Remove the whole inflammatory mass, because simple drainage leads to **nephrocutaneous fistula**.
- Laparoscopic nephrectomy is feasible; conversion rates are high.
- Frozen section can mimic clear cell carcinoma, and XGP can **coexist with RCC** or urothelial carcinoma. If cancer is not excluded, do nephrectomy.

**Diagram to draw:** CT sketch of an enlarged kidney with central staghorn and "bear paw" cavities.

> **Key points to remember:**
>
> - Stone, obstruction, Proteus infection → foamy macrophages.
> - CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
> - Mimics RCC; treat by complete nephrectomy, not drainage.

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# Q3b. Doppler in urology

**Type:** SAQ (10 marks)

## Principle

- **Doppler shift:** sound reflected from moving red cells changes frequency in proportion to velocity and the cosine of the insonation angle: $\Delta f = 2f_0 v \cos\theta / c$. Keep the angle **below 60°**.
- **RI (resistive index) = (PSV − EDV) / PSV** (peak systolic and end-diastolic velocity); reflects downstream resistance.

## Modes

| Mode | Shows | Use |
|---|---|---|
| **Colour** | Velocity and direction (red towards, blue away by convention) | Flow in kidney, testis, ureteric jets |
| **Spectral (duplex)** | Waveform at one point; velocities, RI | Penile and renal arteries |
| **Power** | Amplitude only; 3–5 times more sensitive, less angle-dependent | Low flow: torsion, prostate lesions |

## Applications

| Organ | Findings |
|---|---|
| **Scrotum** | **Torsion:** absent intratesticular flow (compare with the other side); epididymitis: increased flow; **varicocele:** venous reflux on Valsalva; intratesticular RI >0.6 suggests poor spermatogenesis |
| **Penis (after intracavernosal injection)** | **PSV <25 cm/s = arterial ED (erectile dysfunction)**, 25–35 equivocal, >35 normal; **EDV >5 cm/s and RI <0.75 = veno-occlusive dysfunction**; Peyronie plaque assessment |
| **Priapism** | Ischaemic: absent cavernosal flow; non-ischaemic: high flow (PSV >50 cm/s), locates the fistula |
| **Kidney** | Normal RI 0.47–0.70; **raised RI in acute obstruction** (useful in pregnancy, where change in RI is more accurate); renal artery stenosis: high velocity and parvus-tardus waveform; arteriovenous fistula after biopsy or PCNL (percutaneous nephrolithotomy) |
| **Transplant** | Arterial thrombosis: absent signal; **venous thrombosis: reversed diastolic flow**; stenosis: high velocity; rejection: loss of diastolic flow, high RI |
| **Bladder** | Ureteric jets (absent on the obstructed side) |
| **Tumours** | Vascularity of renal and testicular masses; tumour thrombus; hypervascular prostate cancer foci |

- **Also know:** intraoperative micro-Doppler in microsurgical varicocelectomy; contrast-enhanced ultrasound and elastography in multiparametric ultrasound.

## Limitations

- Operator- and angle-dependent; poor in obesity and deep vessels; early torsion may show collateral flow, so **Doppler cannot exclude torsion**.

**Diagram to draw:** spectral waveform marking PSV and EDV with the RI formula.

**Key points to remember:**

- RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
- Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
- Absent intratesticular flow = torsion, but explore on clinical grounds.

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## Q3c. Christian Chaussey

**Type:** SAQ (10 marks)

*Taken as Christian Chaussy, the Munich urologist who pioneered extracorporeal shock wave lithotripsy; answered as the history and principles of SWL.*

### Contribution and history

- Engineers at **Dornier** found by chance that shock waves from raindrop and micrometeorite research affected tissue.
- Chaussy, with Brendel and Schmiedt in **Munich**, developed the technique in animals and performed the **first human SWL (shock wave lithotripsy) on 7 February 1980 on the Dornier HM1**. They reported it in the *Lancet* in 1980.
- **HM3** (water bath, electrohydraulic spark, biplanar fluoroscopy): first installations Stuttgart 1983 and Indianapolis 1984. It remains the reference for efficacy.
- SWL made most open stone surgery obsolete.

### Principles

- A shock wave is a short acoustic pulse: **peak positive 30–100 MPa, then a negative tail of about −10 MPa**.
- Generated at **F1**, focused onto the stone at **F2** through water or gel coupling (water and tissue have similar acoustic impedance); targeting by fluoroscopy or ultrasound.

| Energy source | Feature |
|---|---|
| Electrohydraulic (HM1, HM3) | Underwater spark, ellipsoid reflector; most effective; electrode wear |
| Piezoelectric | Ceramic elements; long life, little pain; small focal zone, low energy |
| **Electromagnetic** (modern) | Coil and membrane with lens; consistent; dominant today |

### Mechanisms of fragmentation

- **Spallation:** reflected wave at the back surface creates tension.
- **Shear and tear** inside the stone.
- **Quasistatic squeezing** and **dynamic squeezing** (strongest force in initial breakage).
- **Cavitation:** collapsing bubbles erode the surface; excess bubbles shield the stone and injure tissue.

### Practical points

- Better results at **60–90 shocks/min**, with power **ramping** and air-free coupling.
- Poor results: stone >1000 HU (Hounsfield units), skin-to-stone distance >10 cm, lower pole >1 cm.
- **Contraindications:** pregnancy, uncorrected coagulopathy, untreated UTI (urinary tract infection), distal obstruction, aortic aneurysm >4 cm.
- **Also know:** burst wave lithotripsy and ultrasonic propulsion are newer descendants.

**Diagram to draw:** ellipsoid reflector with spark at F1 and stone at F2.

> **Key points to remember:**
> - Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1.
> - HM3 (1983) is still the efficacy benchmark.
> - Spallation, squeezing, shear, cavitation; electromagnetic sources today.

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## Q3d. Creatinine clearance

**Type:** SAQ (10 marks) (GFR assessment also answered in 2023-Winter Paper 2, Q3a)

### Definition

- Creatinine clearance (CrCl) is the volume of plasma cleared of creatinine per minute. It estimates **GFR (glomerular filtration rate)** because creatinine is freely filtered and not reabsorbed.
- Normal about 120 mL/min/1.73 $m^2$ in young men, slightly lower in women; falls with age.

### Measured 24-hour creatinine clearance

- **CrCl = (U × V) / P**: U = urine creatinine, V = urine flow (mL/min), P = plasma creatinine.
- **Method:** discard the first morning void, collect all urine for 24 hours including the next morning's first void; take serum creatinine during the collection.
- **Completeness check:** creatinine excretion should be **20–25 mg/kg/day in men, 15–20 mg/kg/day in women**.

### Limitations

- **Tubular secretion** of creatinine: small in health but rising as function falls, so CrCl **overestimates GFR** in CKD (chronic kidney disease).
- Collection errors are common. Creatinine generation depends on **muscle mass**, diet (cooked meat, creatine supplements) and age.
- Drugs blocking secretion (**trimethoprim, cimetidine**) raise serum creatinine without real injury.
- Invalid in AKI (acute kidney injury). **Also know:** cimetidine pretreatment blocks secretion to improve accuracy; averaging urea and creatinine clearance helps in advanced CKD.

### Estimated clearance: Cockcroft–Gault

- **CrCl = (140 − age) × weight (kg) / (72 × serum creatinine mg/dL)**, × 0.85 in women.
- Used for **drug dosing**; not normalised to body surface area; inaccurate in obesity.

### Comparison with other GFR measures

| Method | Comment |
|---|---|
| eGFR (estimated GFR; CKD-EPI [Chronic Kidney Disease Epidemiology Collaboration] 2021) | Age and sex, no race; used for staging |
| Cystatin C | Independent of muscle; combined equation most accurate |
| Measured GFR (inulin, iohexol, $^{51}$Cr-EDTA [ethylenediaminetetraacetic acid], $^{99m}$Tc-DTPA [diethylenetriaminepentaacetic acid]) | Reference standard; DTPA also gives split function |

### Clinical uses

- Drug dosing; **cisplatin eligibility** (CrCl <60 mL/min usually unfit).
- Extremes of body size, amputees and pregnancy, where eGFR is unreliable.
- Living donor evaluation (confirm with measured GFR).

**Diagram to draw:** nephron showing filtration plus a small secreted fraction of creatinine.

**Key points to remember:**

- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

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# Q3e. Radionuclear studies in PUJ obstruction

**Type:** SAQ (10 marks) (also answered in 2021-Summer Paper 1, Q3e)

## Introduction

- Dilatation is not obstruction. **Diuretic renography** is the functional gold standard for PUJ (pelviureteric junction) obstruction and the best follow-up study.

## Tracers

| Tracer | Handling | Use |
|---|---|---|
| $^{99m}$Tc-**MAG3** (mercaptoacetyltriglycine) | Tubular secretion | **Agent of choice**: split function and drainage, good images even with poor function |
| $^{99m}$Tc-**DTPA** (diethylenetriaminepentaacetic acid) | Glomerular filtration | GFR (glomerular filtration rate); poor in renal impairment |
| $^{99m}$Tc-**DMSA** (dimercaptosuccinic acid) | Cortical retention | Static split function and scars; no drainage data |

## "Well-tempered" diuretic renogram

| Element | How | Why |
|---|---|---|
| **Hydration** | Oral, or IV (intravenous) in children | Dehydration mimics obstruction |
| **Bladder catheter** | Urethral catheter | Removes full-bladder and reflux effects |
| **Furosemide** | 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children | Maximal diuresis |
| **Timing** | **F+20** (pelvis full; Campbell authors use F+15), **F0** or **F−15** | Diuretic before filling misleads; always state timing |

## Outputs

- **DRF (differential renal function)** from counts at 1–3 min; normal 45–55%.
- **Drainage half-time (T½)** after furosemide:

| T½ | Interpretation |
|---|---|
| <10 min | Not obstructed |
| 10–20 min | Equivocal |
| >20 min | Obstructed |

- **O'Reilly curves:** I normal; II obstructed (keeps rising); IIIa dilated, non-obstructed (falls after furosemide); IIIb equivocal.
- **Also know:** P40 (activity left at 40 min) raises sensitivity; biphasic curve and poor upright drainage predict progression.

## Indications for pyeloplasty

- **DRF <40%**, **fall in DRF** on serial comparable scans (≥5% in children; 10% in older series), poor drainage with symptoms, infection or increasing dilatation.
- Adult DRF <15%: likely non-salvageable; nephrostomy and reassess, or nephrectomy.

## Pitfalls

- Dehydration, early diuretic, full bladder, very poor function.

- No agreed paediatric T½ cut-off (false positives up to 44%); one Campbell chapter uses **T½ >40 min** in adults. Serial trends decide.

## Follow-up and alternatives

- After pyeloplasty: renogram 1 month after stent removal. Equivocal cases: **Whitaker test** or MRU (magnetic resonance urography).

**Diagram to draw:** O'Reilly curves I, II, IIIa and IIIb.

> **Key points to remember:**
> - MAG3 tracer of choice; hydrate and catheterise.
> - DRF normal 45–55%; T½ <10 normal, >20 min obstructed.
> - Operate for DRF <40%, falling DRF, or symptoms.

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# Q3f. Cyclosporins

**Type:** SAQ (10 marks) (immunosuppressants also answered in 2021-Summer Paper 1, Q3f; tacrolimus in 2017-Winter Paper 3, Q3c)

*Taken as ciclosporin (cyclosporin A).*

## Introduction

- Ciclosporin is a cyclic peptide of 11 amino acids from the fungus *Tolypocladium inflatum*. It was the first **CNI (calcineurin inhibitor)**.
- Its introduction in the early 1980s cut acute rejection from at least 60% (pre-ciclosporin era) towards today's <10%. Tacrolimus has now largely replaced it.

## Mechanism

```
Ciclosporin + cyclophilin (cytoplasm)
   → complex inhibits calcineurin
   → NFAT (nuclear factor of activated T cells) stays phosphorylated
   → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)
```

## Pharmacokinetics and monitoring

- Variable oral absorption; the **microemulsion form** absorbs more predictably.
- Metabolised by **CYP3A4** (cytochrome P450 3A4) and P-glycoprotein; narrow therapeutic index, so whole-blood **trough (C0) or 2-hour (C2) levels** guide dosing.

## Uses

- **Transplantation:** maintenance with an antimetabolite and steroids, when tacrolimus is not tolerated.
- **IC/BPS (interstitial cystitis/bladder pain syndrome):** refractory **Hunner lesions**, 1–3 mg/kg/day; better than pentosan polysulfate.
- **Others:** retroperitoneal fibrosis after steroid failure; frequently relapsing nephrotic syndrome; severe psoriasis and other dermatoses.

## Toxicity

| System | Effect |
|---|---|
| Kidney | **Nephrotoxicity**: acute afferent arteriolar vasoconstriction (reversible); chronic striped fibrosis, arteriolar hyalinosis; TMA (thrombotic microangiopathy) |
| Metabolic | Hypertension, hyperkalaemia, hypomagnesaemia, **hyperuricaemia** (gout), dyslipidaemia |
| Cosmetic | **Gingival hyperplasia, hirsutism** |
| Neurological | Tremor, PRES (posterior reversible encephalopathy syndrome) |
| Immune | Infection, skin cancer, PTLD (post-transplant lymphoproliferative disorder) |

## Interactions

- **Levels rise:** azoles, erythromycin and clarithromycin, diltiazem, verapamil, grapefruit.
- **Levels fall:** rifampicin, phenytoin, carbamazepine, St John's wort.
- More nephrotoxicity with NSAIDs (non-steroidal anti-inflammatory drugs) and aminoglycosides.

## Versus tacrolimus

- Ciclosporin: more hirsutism, gum hypertrophy, hypertension and hyperlipidaemia; less diabetes and tremor.

- **ELITE-Symphony (2007):** low-dose tacrolimus gave less rejection and better GFR (glomerular filtration rate) than ciclosporin regimens.
- **Urological note:** CNIs promote transplant stones (hyperuricaemia, hyperoxaluria, hypocitraturia).

**Diagram to draw:** T cell with the cyclophilin–calcineurin–NFAT–IL-2 pathway.

> **Key points to remember:**
> - Cyclophilin → calcineurin → NFAT → IL-2 blocked.
> - Nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
> - Urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

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# Rapid Revision Sheet – Practice Set 4 Paper 1

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**Q1. Discuss pathophysiology of renal stones and also describe how to prevent its recurrence (LAQ)**

- Metastable zone between Ksp and Kf: where inhibitors and therapy act.
- Randall plaque anchors idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi).
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.

**Q2. Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management (LAQ)**

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- BT and PFMT with urge suppression first: 50–80% fewer leaks.
- PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

**Q3a. Xanthogranulomatous pyelonephritis. (SAQ)**

- Stone, obstruction, Proteus infection → foamy macrophages.
- CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
- Mimics RCC; treat by complete nephrectomy, not drainage.

**Q3b. Doppler in urology (SAQ)**

- RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
- Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
- Absent intratesticular flow = torsion, but explore on clinical grounds.

**Q3c. Christian Chaussey (SAQ)**

- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1.
- HM3 (1983) is still the efficacy benchmark.
- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

**Q3d. Creatinine clearance (SAQ)**

- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

**Q3e. Radionuclear studies in PUJ obstruction (SAQ)**

- MAG3 tracer of choice; hydrate and catheterise.
- DRF normal 45–55%; T½ <10 normal, >20 min obstructed.
- Operate for DRF <40%, falling DRF, or symptoms.

**Q3f. Cyclosporins (SAQ)**

- Cyclophilin → calcineurin → NFAT → IL-2 blocked.
- Nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.